

⊗ Find the first 6 pixels for:  $(3, 7)$  to  $(10, -17)$   
 Let,  $P_1$   $P_2$

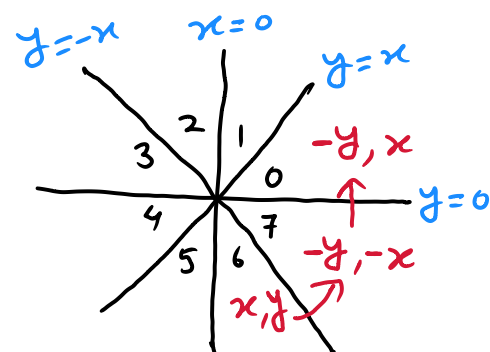
⇒ i) Find Zone

$$\left. \begin{aligned} dx &= 10 - 3 = 7 (+) \\ dy &= -17 - 7 = -24 (-) \end{aligned} \right\} \begin{array}{c|c} -+ & ++ \\ \hline -- & +- \end{array}$$

$$\left. |dx| < |dy| \rightarrow \text{Zone 6} \right\} \begin{array}{c} \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ \hline \diagup \quad \diagdown \\ \diagdown \quad \diagup \end{array}$$

$dy > dx$

ii) Convert Zone 6 to Zone 0



Zone 6 to Zone 0

$$\begin{aligned} x &= -y \\ y &= x \end{aligned}$$

So,

$$\begin{aligned} \text{Zone 6} &\rightarrow \text{Zone 0} \\ P_1(3, 7) &\rightarrow P'_1(-7, 3) \\ P_2(10, -17) &\rightarrow P'_2(17, 10) \end{aligned}$$

iii) Midpoint Line Drawing Algo. for Zone 0

$$dx = 17 + 7 = 24$$

$$dy = 10 - 3 = 7$$

$$d_{init} = 2dy - dx = 14 - 24 = -10$$

$$\Delta d_{NE} = 2dy - 2dx = 14 - 48 = -34$$

$$\Delta d_E = 2dy = 14$$

$$d_{init} = -10; \Delta d_{NE} = -34; \Delta d_E = 14$$

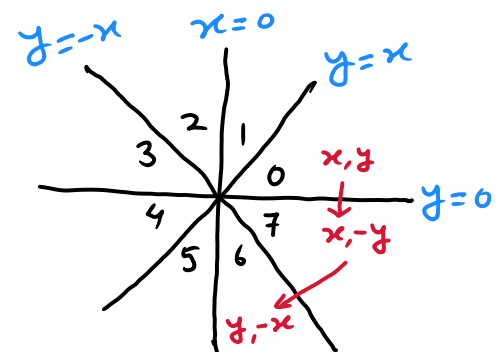
| $x$ | $y$ | $d$       | NE/E | $d_{update}$              | Pixel     |
|-----|-----|-----------|------|---------------------------|-----------|
| -7  | 3   | $-10 < 0$ | E    | $-10 + \Delta d_E = 4$    | $(-7, 3)$ |
| -6  | 3   | $4 > 0$   | NE   | $4 + \Delta d_{NE} = -30$ | $(-6, 3)$ |
| -5  | 4   | $-30 < 0$ | E    | $-30 + \Delta d_E = -16$  | $(-5, 4)$ |
| -4  | 4   | $-16 < 0$ | E    | $-16 + \Delta d_E = -2$   | $(-4, 4)$ |
| -3  | 4   | $-2 < 0$  | E    | $-2 + \Delta d_E = 12$    | $(-3, 4)$ |
| -2  | 4   |           |      |                           | $(-2, 4)$ |

Pixels of Zone 0

iv) Convert Zone 0 to Zone 6

| <u>Zone 0</u> | $\longrightarrow$ | <u>Zone 6</u> |
|---------------|-------------------|---------------|
| -7, 3         |                   | 3, 7          |
| -6, 3         |                   | 3, 6          |
| -5, 4         |                   | 4, 5          |
| -4, 4         |                   | 4, 4          |
| -3, 4         |                   | 4, 3          |
| -2, 4         |                   | 4, 2          |

Pixels of Zone 6



Zone 6 to Zone 0

$$x = y$$

$$y = -x$$

(Answer)

## Derived Formulas:

$$\frac{\text{Zone 1} \rightarrow \text{Zone 0}}{x = Y, y = X}$$

$$\frac{\text{Zone 0} \rightarrow \text{Zone 1}}{x = Y, y = X}$$

$$\frac{\text{Zone 2} \rightarrow \text{Zone 0}}{x = Y, y = -X}$$

$$\frac{\text{Zone 0} \rightarrow \text{Zone 2}}{x = -Y, y = X}$$

$$\frac{\text{Zone 3} \rightarrow \text{Zone 0}}{x = -X, y = Y}$$

$$\frac{\text{Zone 0} \rightarrow \text{Zone 3}}{x = -X, y = Y}$$

$$\frac{\text{Zone 4} \rightarrow \text{Zone 0}}{x = -X, y = -Y}$$

$$\frac{\text{Zone 0} \rightarrow \text{Zone 4}}{x = -X, y = -Y}$$

$$\frac{\text{Zone 5} \rightarrow \text{Zone 0}}{x = -Y, y = -X}$$

$$\frac{\text{Zone 0} \rightarrow \text{Zone 5}}{x = -Y, y = -X}$$

$$\frac{\text{Zone 6} \rightarrow \text{Zone 0}}{x = -Y, y = X}$$

$$\frac{\text{Zone 0} \rightarrow \text{Zone 6}}{x = Y, y = -X}$$

$$\frac{\text{Zone 7} \rightarrow \text{Zone 0}}{x = X, y = -Y}$$

$$\frac{\text{Zone 0} \rightarrow \text{Zone 7}}{x = X, y = -Y}$$